SENTINEL-GNSS — Individual Poster

Architecture, Training, EKF Integration, and Documentation

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My Role in the Project

I led the core technical pipeline of SENTINEL-GNSS:

- Repository creation and code architecture
- 37-feature specification (`feature_list.py`)
- Transformer+BiLSTM neural network design & training
- 9-state EKF formulation (Joseph-form covariance)
- SENTINEL adaptive R integration
- Huber robust EKF (IRLS) — calibrated for each receiver
- Bootstrap Particle Filter with Student-t likelihood
- Inference packaging (0.039 ms/sample)
- Dashboard backend + Huber/PF track integration
- All documentation and paper writing (3 docs, 5 paper sections)

Architecture I Designed

Input: $(B, 60, 37)$ feature windows

Transformer Encoder

2 layers · 4 heads · $d = 128$ · $d_{ff} = 512$

Captures: PDOP trends, satellite rise/set events

BiLSTM

2 layers · hidden 256 (128/direction)

Captures: sequential C/N_0 decline, directional approach

3 Output Heads: +5 s, +15 s, +30 s (one forward pass, 1.46 M params)

Training: Focal loss $\gamma = 1.0$, class weights $[1, 2, 5]$, AdamW 10^{-3}, cosine annealing, early stopping patience 20

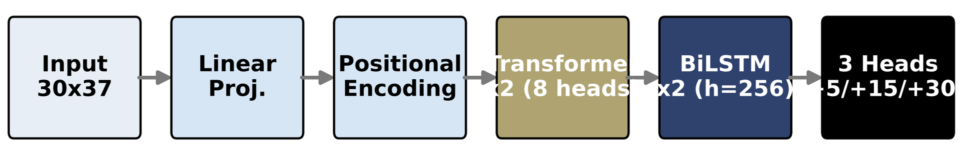

Figure 1: SENTINEL-GNSS hybrid architecture.

Why Hybrid? (Ablation)

Architecture	Macro-F1	DEG F1
Transformer-only	0.767	0.571
BiLSTM-only	0.767	0.645
Full (mine)	**0.821**	**0.718**

Transformer alone over-flags (precision 0.42). BiLSTM alone under-recalls (recall 0.645 vs **0.847**). Together they are complementary.

SENTINEL Adaptive EKF

9-state EKF: $\mathbf{x} = [x, y, v_x, v_y, \psi, b, b_{a_x}, b_{a_y}]^\top$

At each epoch, SENTINEL P(DEGRADED) selects R:

$$\mathbf{R} = \begin{cases} (40\,\mathrm{m})^2\mathbf{I} & P > 0.45 \text{ (pre-emptive)} \\ r_{\text{base}}^2\mathbf{I} & \text{otherwise} \end{cases}$$

Kalman gain reduces **before** bad GPS arrives. Joseph-form update for numerical stability.

Huber Robust EKF (My Design)

Addresses non-Gaussian GPS noise (NLOS heavy tails):

$$w_i = \min\left(1, \frac{c\,\tilde{\sigma}_i}{|\tilde{y}_i|}\right), \quad R_{ii}^H = \frac{R_{ii}}{w_i}$$

Key calibration insight (my finding):
Threshold $= c \times r_{\text{base}}$ must match ACTUAL GPS error scale.

Receiver	c	Threshold
Trimble	5.0	20 m
u-blox	30.0	240 m

u-blox 99th-pct innovation: 208.9 m.
$c = 5$ (threshold 40 m) fired on 22.7 % of valid updates.
$c = 30$ fires only on the 1472 m outlier spike.

SENTINEL + Huber conflict: Must run `adaptive=False`.
Stacking both: SENTINEL lowers gain → drift → large innovations → Huber suppresses recovery → filter paralysis.

Particle Filter (My Design)

N = 500 particles · Student-t GPS likelihood ($\nu = 3$):

$$\log p \propto -\tfrac{\nu+1}{2}\left[\log\left(1 + \tfrac{d_x^2}{\nu r^2}\right) + \log\left(1 + \tfrac{d_y^2}{\nu r^2}\right)\right]$$

ESS threshold N/3 · NHC $r_{\text{nhc}} = 1.0$ m/s
Jitter: $[3.0, 3.0, 0.2, 0.2, 0.05, 0.2, 0.02, 0.02]$ m

GPS attractor phenomenon (my discovery): With persistent bias & N=500, weight ratio climbs $1.65^5 \approx 13$ over 5 epochs → all particles cluster at biased GPS. Needs N2000 or multi-hypothesis proposal to fix.

Fusion Results

Method / Scenario	Deg.RMSE	Gain
EKF Fixed-R · Trimble	**24.3 m**	**48.8 %**
Huber EKF · u-blox	**58.6 m**	**25.2 %**
PF Student-t · Odaiba	**35.4 m**	**40.1 %**

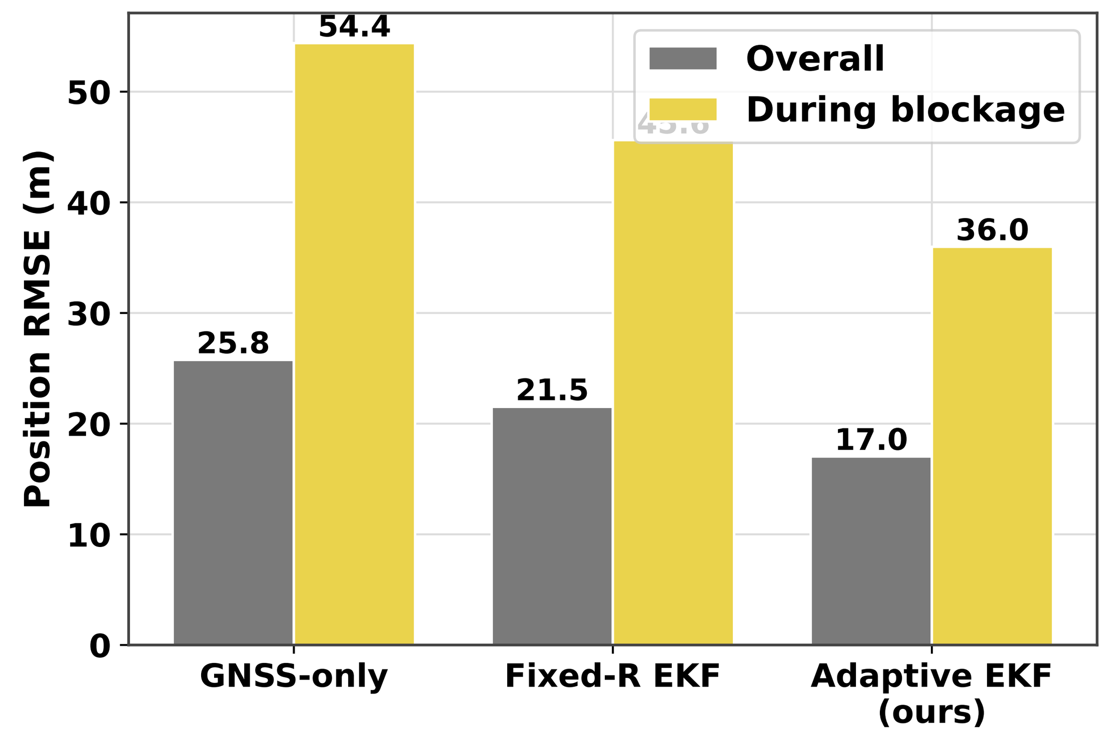

Figure 2: Degraded-segment RMSE across all methods.

The "Swirling" Problem Solved

Without SENTINEL, the EKF swirls between successive NLOS GPS fixes (each pulls the estimate off-course).

With SENTINEL pre-emptive R inflation: Kalman gain drops *before* bad GPS arrives. IMU dead-reckoning dominates the 5–30 s NLOS window. Trajectory stays smooth and accurate.

The vehicle's downstream controller gets a **reliable position** to act on — which is the whole point.

Feature Saliency (My Interpretation)

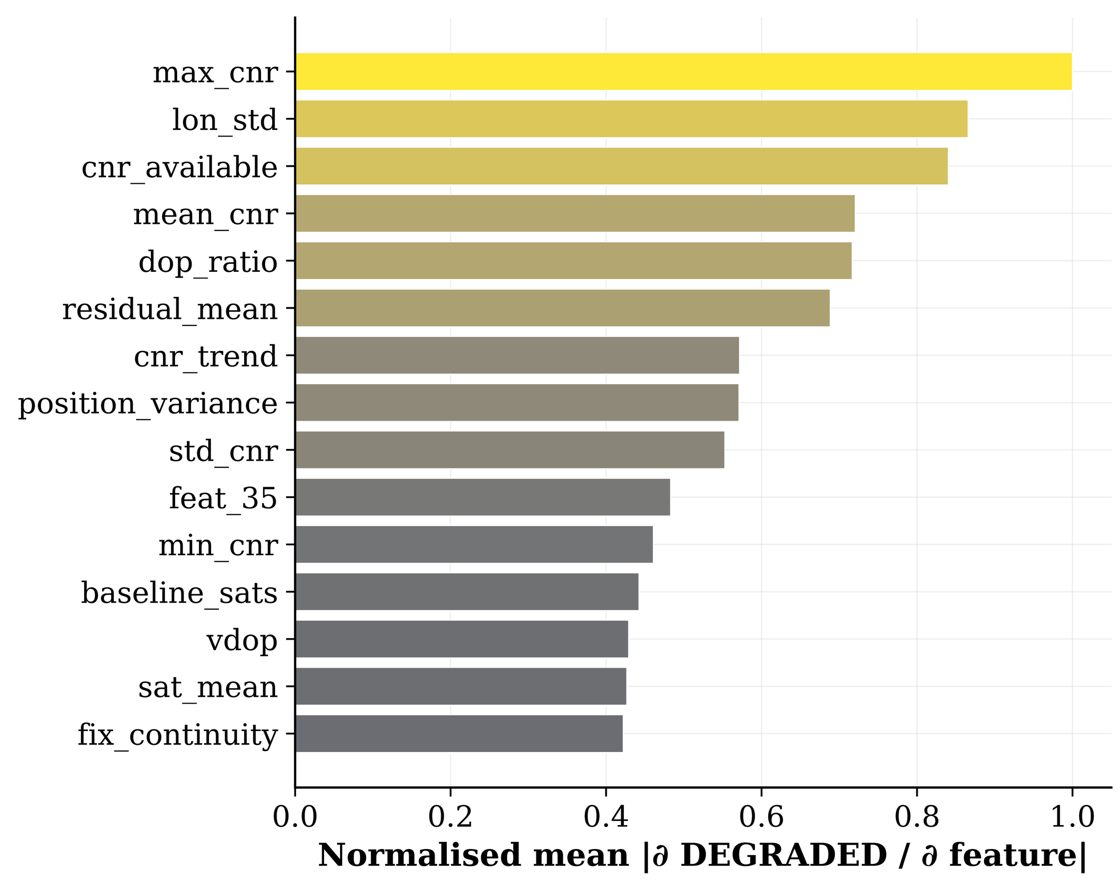

Figure 3: Most important features for +5 s prediction. PDOP, pseudorange residuals, and $\Delta C/N_0$ are top-ranked.

Key Lesson

Calibrate Huber c to actual GPS errors, not model parameters. Setting $c = 5$ with $r_{\text{base}} = 8$ m gives 40 m threshold — fires on 22.7 % of valid u-blox epochs, causing drift. Setting $c = 30$ gives 240 m threshold — fires only on the catastrophic outlier. Result: 46.6 m RMSE vs 88 m with wrong c.